EXPERIMENT 3

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1.Command Line Interface:

PYTHON CODE:

```
import sys


if len(sys.argv) != 2:
    print("Usage: python wordcount.py <filename>")
    sys.exit(1)


filename = sys.argv[1]


try:
    with open(filename, 'r', encoding='utf-16') as file:
        lines = 0
        words = 0
        characters = 0


        for line in file:
            stripped_line = line.strip()


            if stripped_line != "":
                lines += 1


            words += len(stripped_line.split())
            characters += len(stripped_line)
```

```
    print("----- File Statistics -----")
    print("Non-empty Lines :", lines)
    print("Words          :", words)
    print("Characters     :", characters)

except FileNotFoundError:
    print("Error: File not found.")
```

OUTPUT:

```
(base) PS C:\Users\nisha> cd "C:\Users\nisha\OneDrive\Desktop\UID"
>>
(base) PS C:\Users\nisha\OneDrive\Desktop\UID> py wordcount.py file.txt
----- File Statistics -----
Non-empty Lines : 2
Words           : 2
Characters      : 10
```

2. Graphical User Interface (GUI)

Python Code

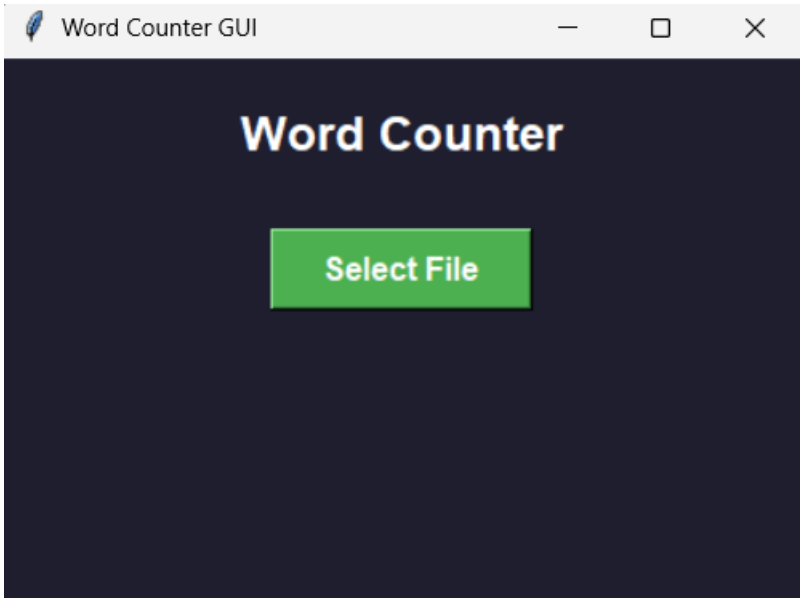
```
import tkinter as tk
from tkinter import filedialog, messagebox

def count_words():
    file_path = filedialog.askopenfilename()

    if file_path:
        try:
            with open(file_path, 'r', encoding='utf-16') as file:
                content = file.read()

            lines = len([line for line in content.splitlines() if line.strip() != ""])
            words = len(content.split())
```

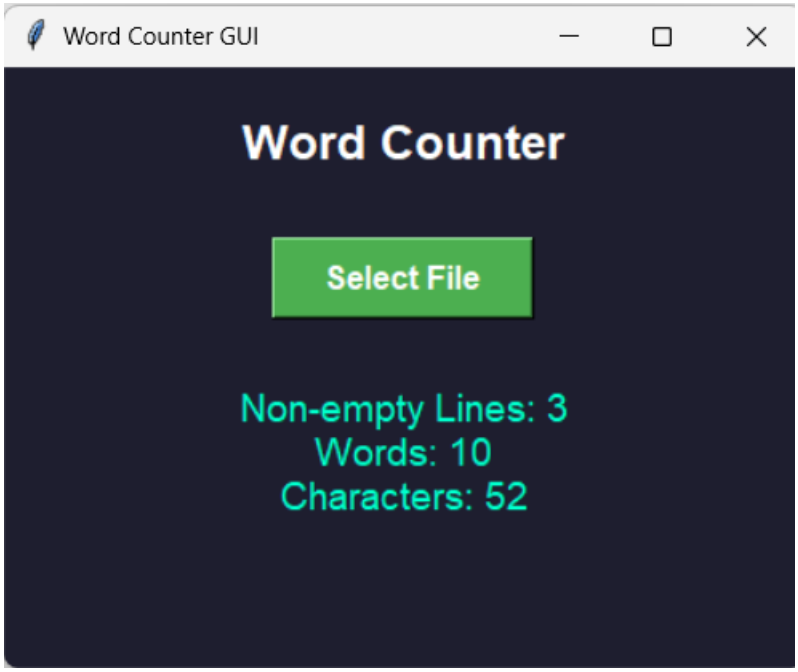
```
        characters = len(content.replace("\n", "").replace("\r", ""))

        result_label.config(
            text=f"Non-empty Lines: {lines}\nWords: {words}\nCharacters: {characters}"
        )

    except Exception as e:
        messagebox.showerror("Error", str(e))

# Main Window
root = tk.Tk()
root.title("Word Counter GUI")
root.geometry("400x300")
root.configure(bg="#1e1e2f")  # Dark background

# Title Label
title = tk.Label(
    root,
    text="Word Counter",
    font=("Helvetica", 18, "bold"),
    bg="#1e1e2f",
    fg="white"
)
title.pack(pady=20)

# Styled Button
btn = tk.Button(
    root,
```

```
    text="Select File",
    command=count_words,
    font=("Helvetica", 12, "bold"),
    bg="#4CAF50",
    fg="white",
    activebackground="#45a049",
    padx=20,
    pady=5
)
btn.pack(pady=10)

# Result Display
result_label = tk.Label(
    root,
    text="",
    font=("Helvetica", 14),
    bg="#1e1e2f",
    fg="#00ffcc",
    justify="center"
)
result_label.pack(pady=20)

root.mainloop()
```

OUTPUT:

```
(base) PS C:\Users\nisha> cd "C:\Users\nisha\OneDrive\Desktop\UID"
(base) PS C:\Users\nisha\OneDrive\Desktop\UID> py wordcount_gui.py
```





3. Voice User Interface (VUI)

PYTHON CODE:

```
import speech_recognition as sr

import os


def rename_file_from_voice_command(command):

    try:
```

```
        words = command.lower().split(" ")

        if "rename" in words and "to" in words:
            old_name = words[words.index("rename") + 1]
            new_name = words[words.index("to") + 1]

            if not os.path.exists(old_name):
                print(f"Error: File '{old_name}' not found.")
                return

            os.rename(old_name, new_name)
            print(f"File renamed from '{old_name}' to '{new_name}'")
        else:
            print("Invalid command format.")
    except Exception as e:
        print(f"Error: {e}")

def listen_for_command():
    recognizer = sr.Recognizer()
    mic = sr.Microphone()

    print("Listening for command...")
    with mic as source:
        recognizer.adjust_for_ambient_noise(source)
        audio = recognizer.listen(source)

    try:
        command = recognizer.recognize_google(audio)
```

```
        print(f"Command received: {command}")
        rename_file_from_voice_command(command)
    except sr.UnknownValueError:
        print("Could not understand the command.")
    except sr.RequestError as e:
        print(f"Service error: {e}")


if __name__ == "__main__":
    listen_for_command()
```

OUTPUT:

```
(base) PS C:\Users\nisha\OneDrive\Desktop\vs code>   C:\Users\nisha\miniconda3\python.exe voice_rename.py
Listening for command to rename a file...
Command received: rename sample.txt to file.txt
File renamed from sample.txt to file.txt
(base) PS C:\Users\nisha\OneDrive\Desktop\vs code>
```

4. User Satisfaction Comparison

PYTHON CODE:

```
def survey():
    print("Rate your satisfaction with the following interfaces (1-5):")


    try:
        cli = int(input("CLI (1-5): "))
        gui = int(input("GUI (1-5): "))
        vui = int(input("VUI (1-5): "))


        if not (1 <= cli <= 5 and 1 <= gui <= 5 and 1 <= vui <= 5):
            print("Please enter values between 1 and 5.")
            return


        print("\nSatisfaction Ratings:")
```

```
        print(f"CLI: {cli}")
        print(f"GUI: {gui}")
        print(f"VUI: {vui}")

        avg = (cli + gui + vui) / 3
        print(f"\nAverage Satisfaction Score: {avg:.2f}")

    except ValueError:
        print("Invalid input! Enter numeric values.")

if __name__ == "__main__":
    survey()
```

OUTPUT:

```
(base) PS C:\Users\nisha\OneDrive\Desktop\vs code> python user_satisfaction.py
>>
Rate your satisfaction with the following interfaces (1-5):
CLI (1-5): 4
GUI (1-5): 5
VUI (1-5): 4

Your satisfaction ratings:
CLI: 4
GUI: 5
VUI: 4

Average Satisfaction Score: 4.33
(base) PS C:\Users\nisha\OneDrive\Desktop\vs code>
```